Evaluation of Control Methods for a Temperature Control Apparatus

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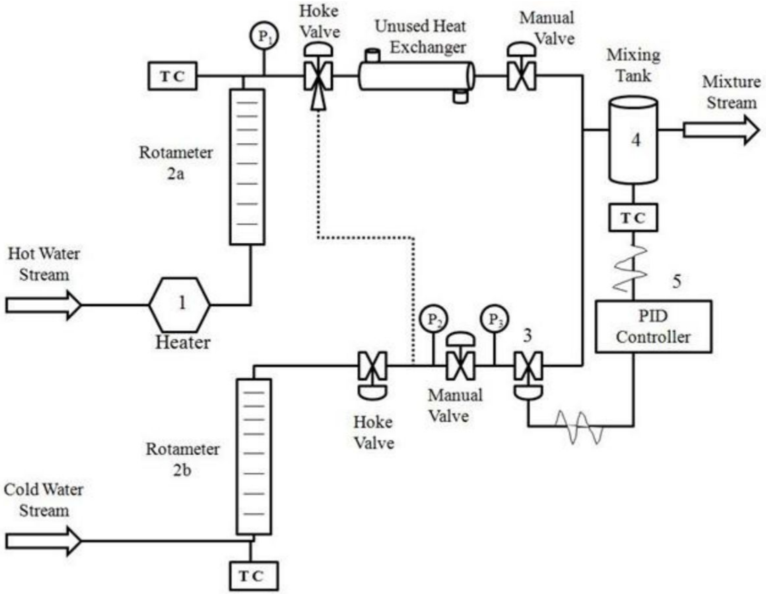
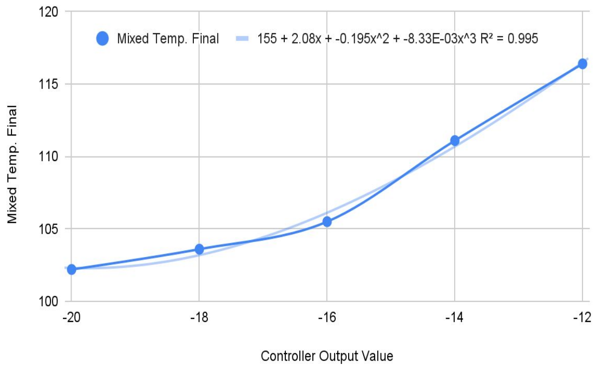
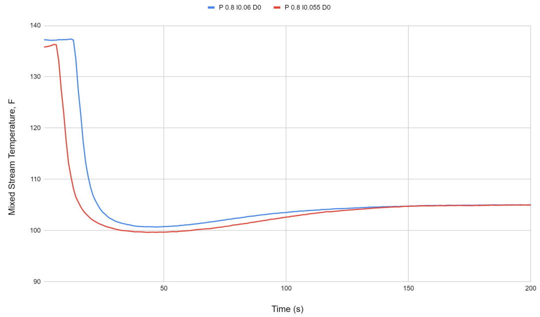


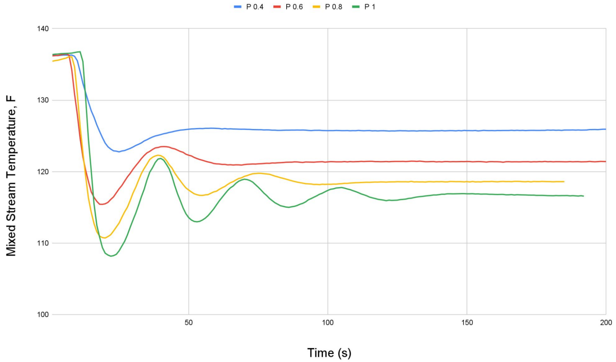
This presentation focuses on



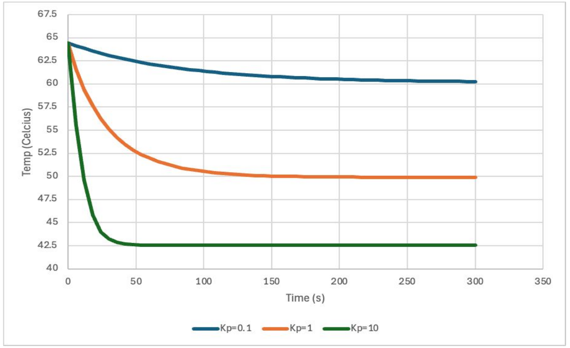
Manual Control



Fine-Tuning

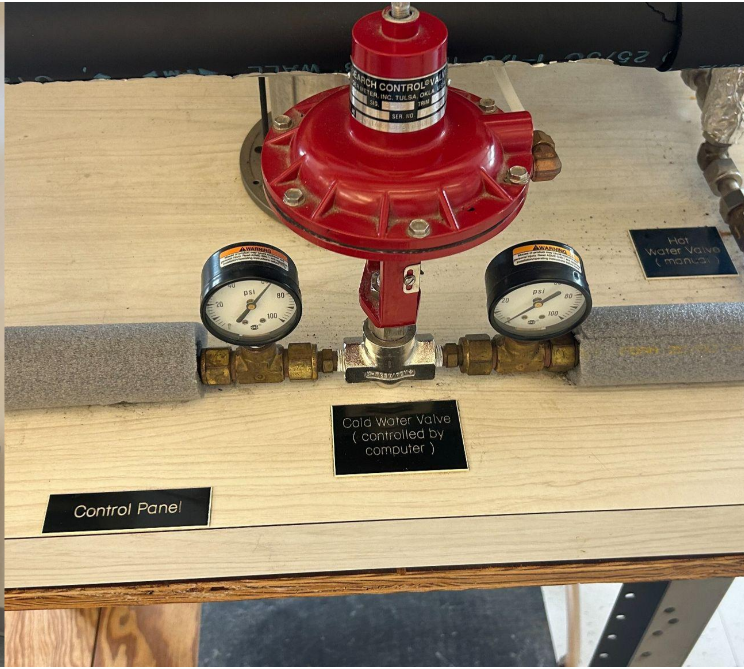


Automatic Control



Analysis of controller parameters

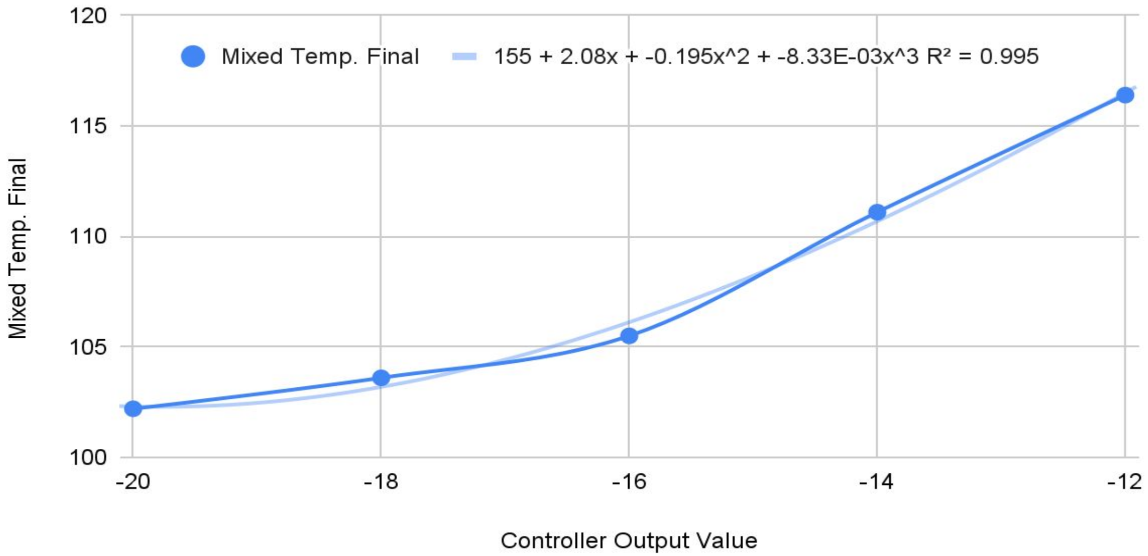
The experimental apparatus is made of multiple control valves, temperature thermocouples, and a mixing chamber.

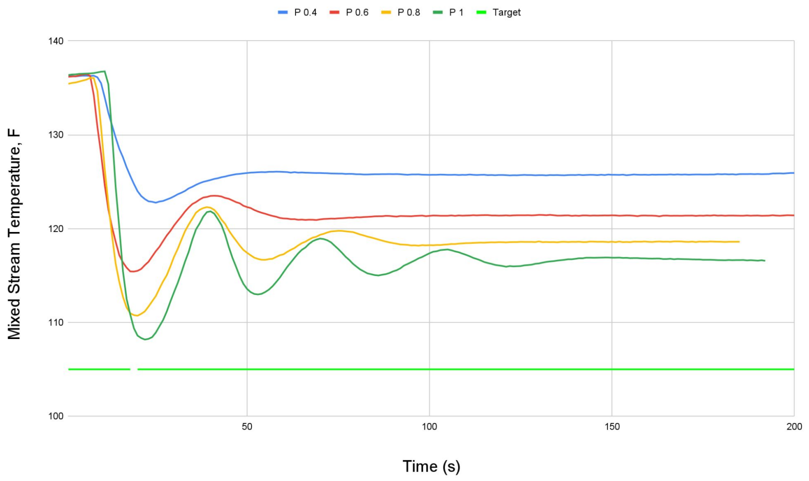


Percent discrepancy between $T_{mix, theory}$ and $T_{mix, experimental}$ showed that manual trials were more precise

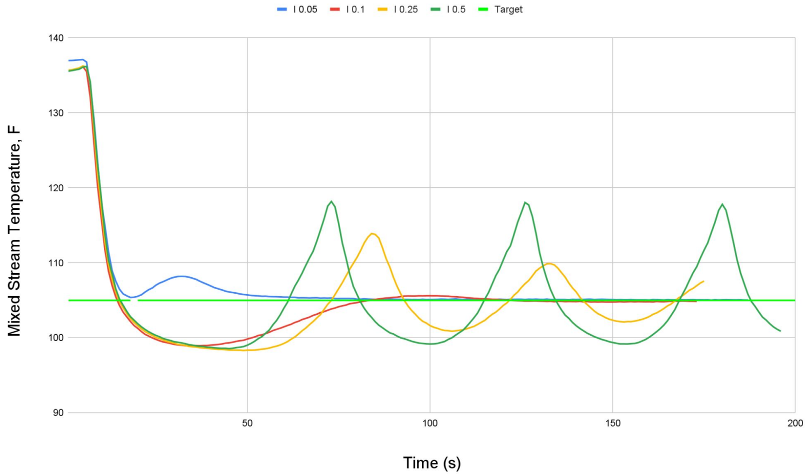
Method	$T_{mix, theory (avg)}$	$T_{mix, experimental (avg)}$	% Difference (avg)
Manual	108.93 ± 5.84	107.76	1.20%
Automatic	113.18 ± 6.70	108.75	4.91%

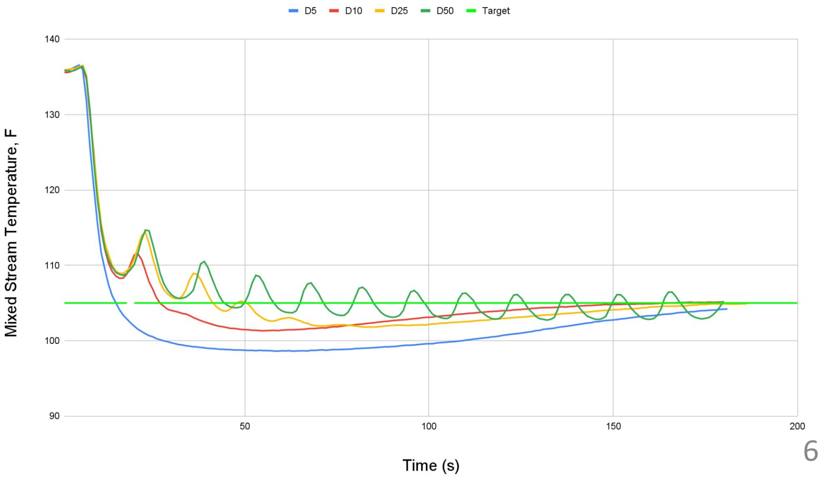
Low difference between the $T_{mix,\ calculated}$ and $T_{mix,\ theoretical}$ showed that manual control is a viable control method



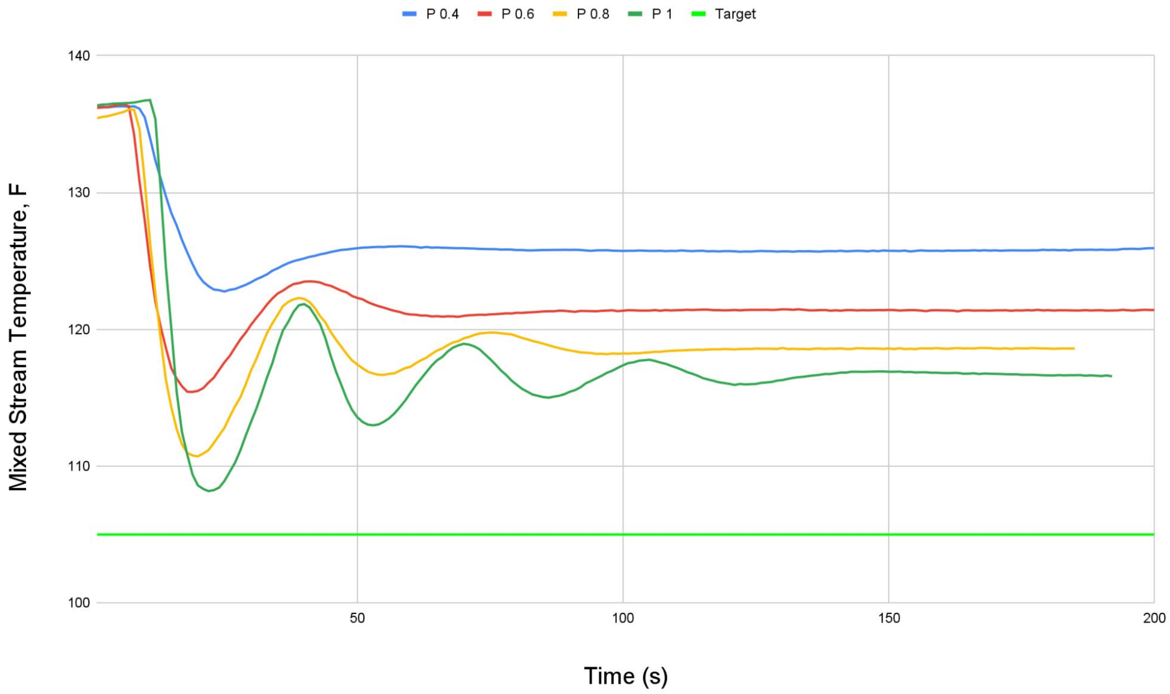


It is important to determine how each parameter influences the final mixing temperature, before the system can be optimized



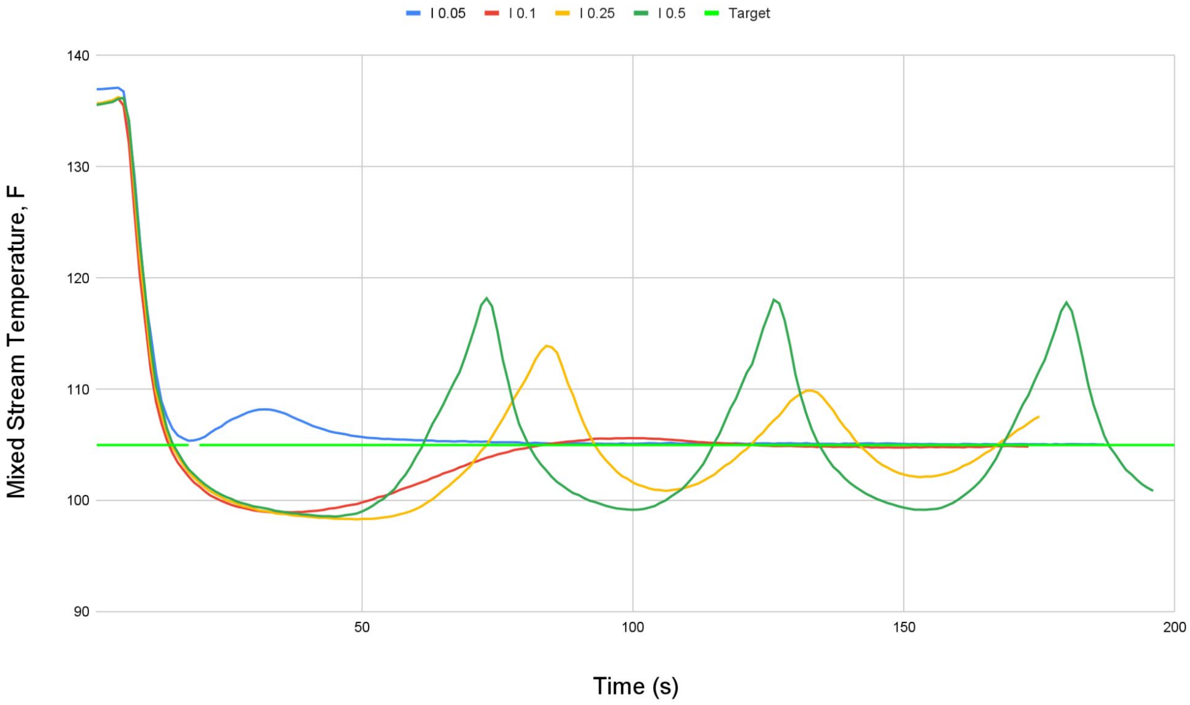


Adjusting K_P changes final temperature at low values but introduces oscillatory patterns at higher values



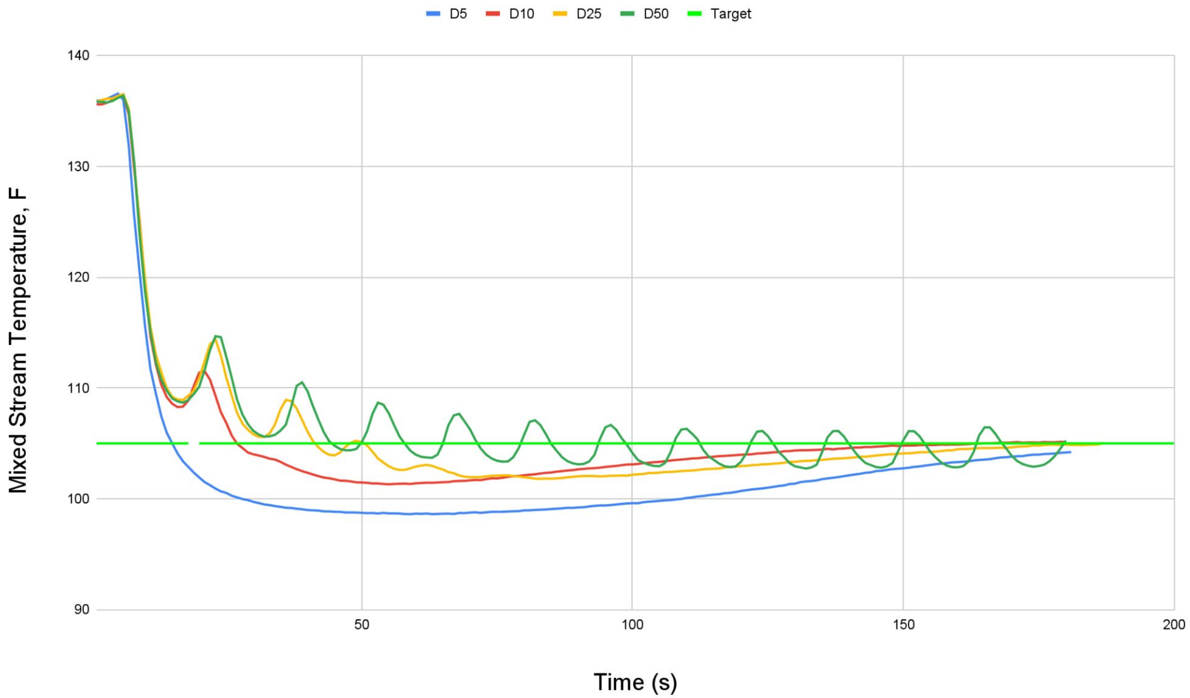
Optimal K_P : 0.8

Adjusting K_I corrects K_P offset to target temperature, but causes overshoots and overcorrects at higher values



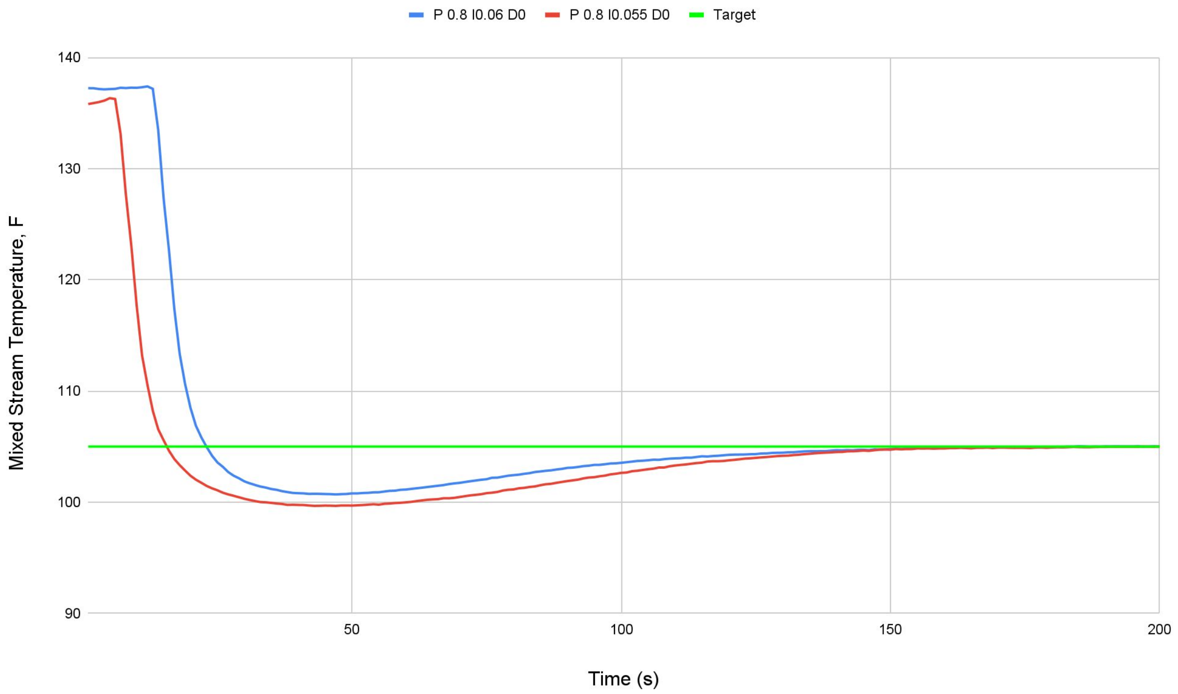
Optimal K_I : 0.055

Adjusting K_D increases time to target temperature at low values and introduces wave-like function at high values

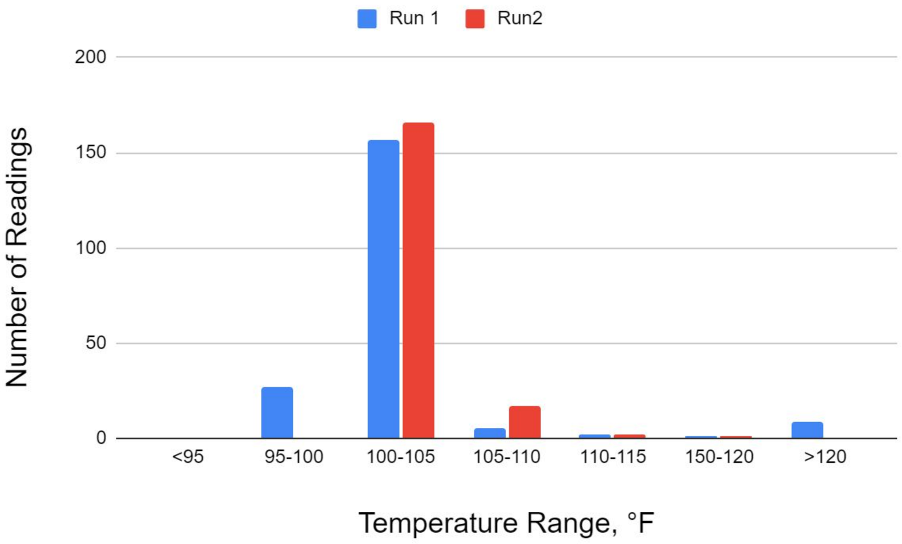


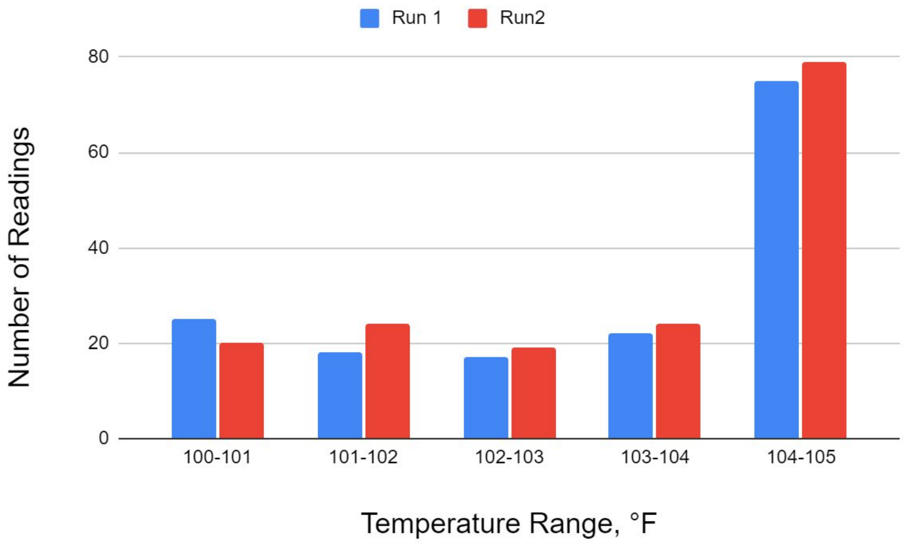
Optimal K_D : 0

By fine-tuning, we determined that the optimal settings to decrease the final mixing temperature to 105°C were P: 0.8, I: 0.055, and D: 0

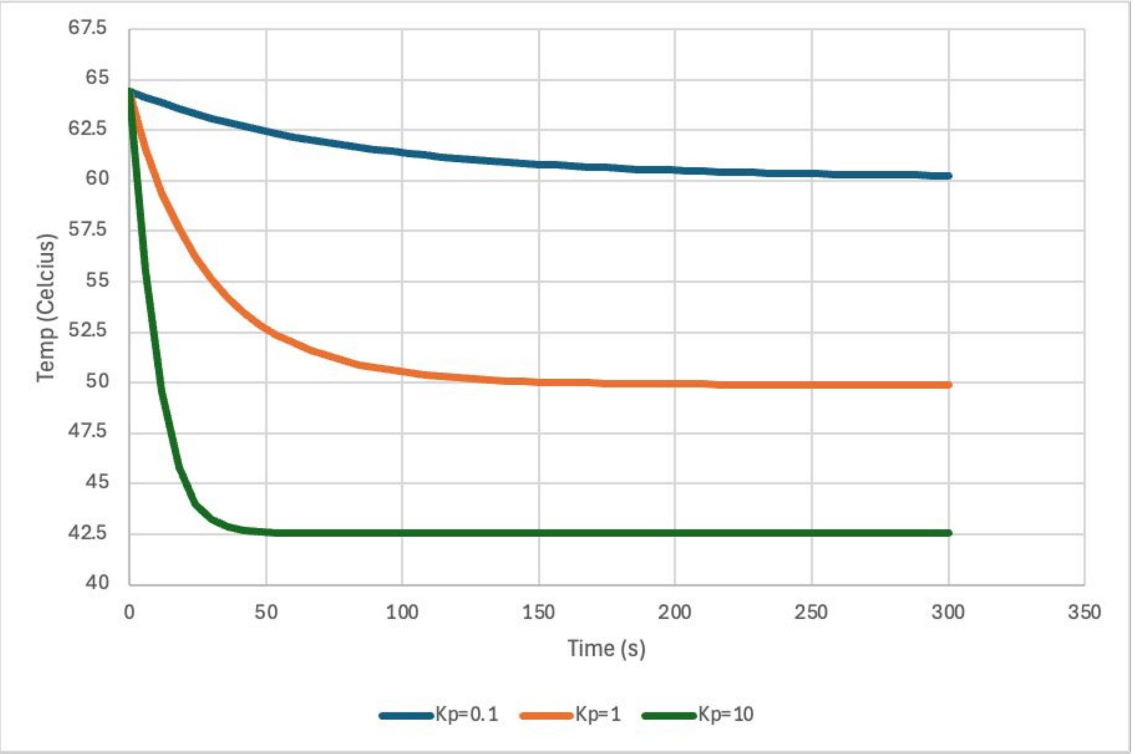


Looking at the frequency of each run at varying temperatures, we reinforced that Run 2, where P:0.8, I:0.055, D:0.0, is the better setting for achieving 105°C.



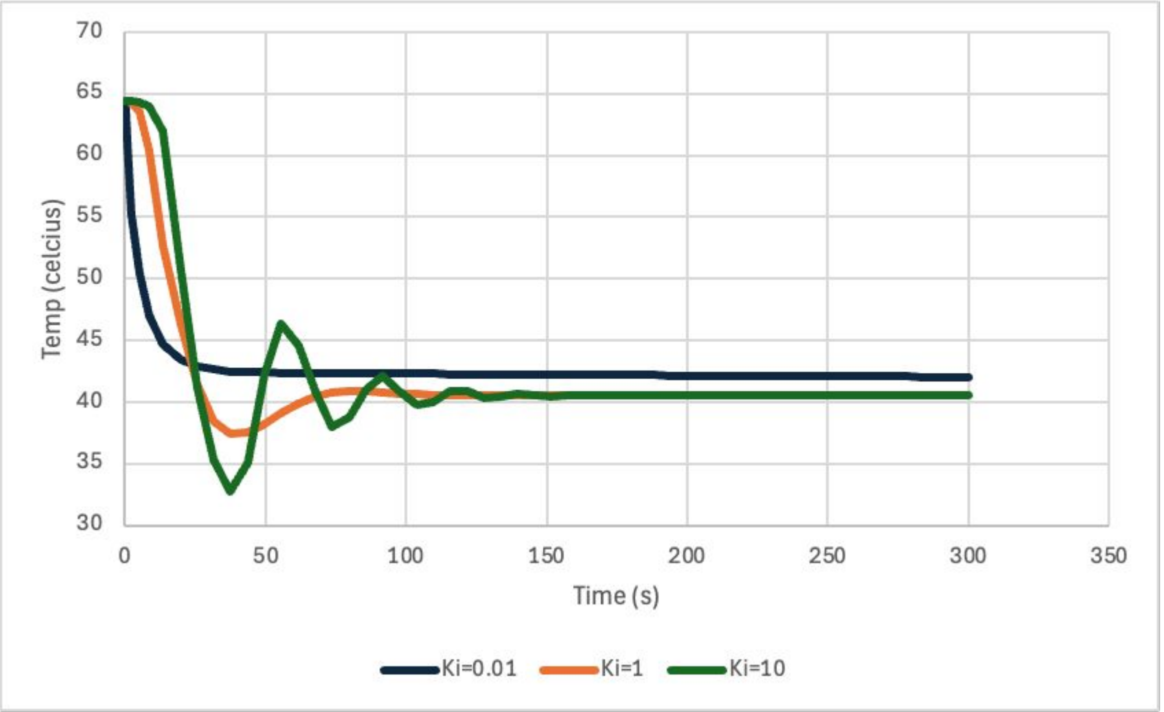


Through a MATLAB simulation we determined that larger K_p values reduce offset from the starting point



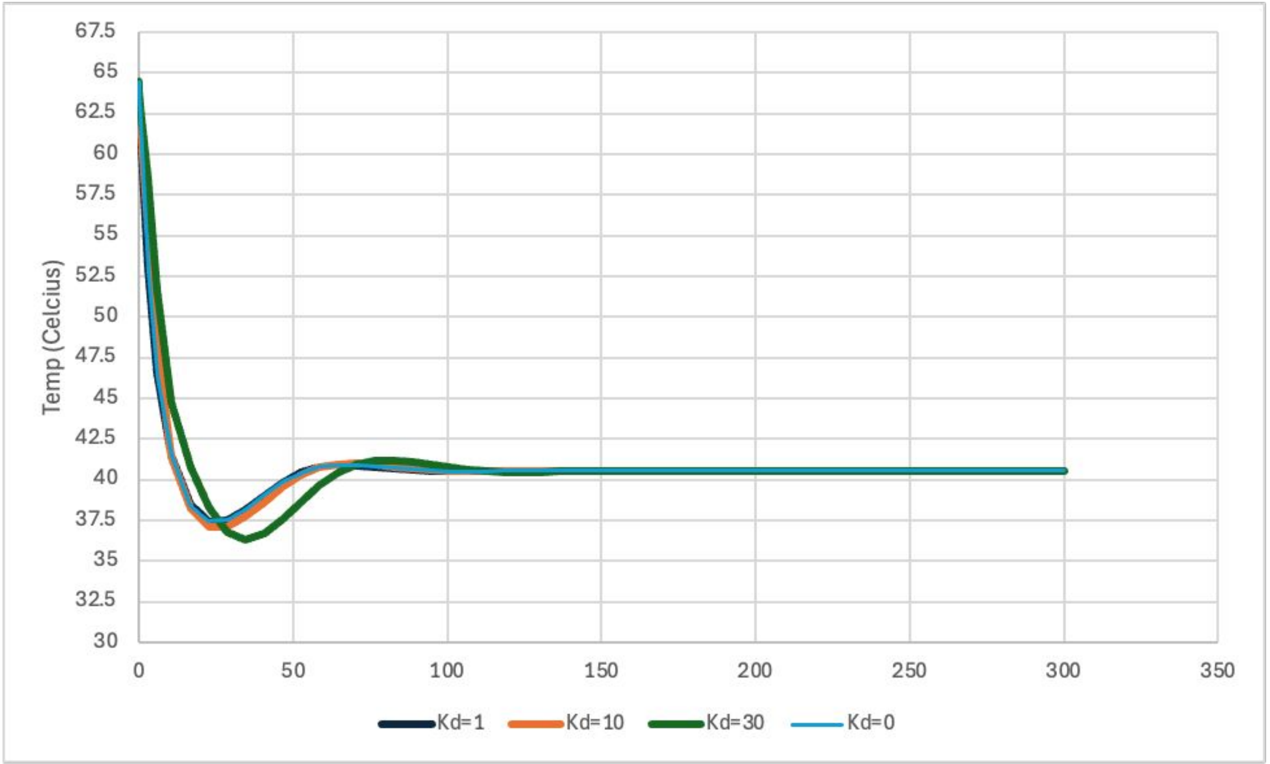
Optimal K_p : 10

The MATLAB simulation showed that lower K_I values reduce oscillations around the setpoint.



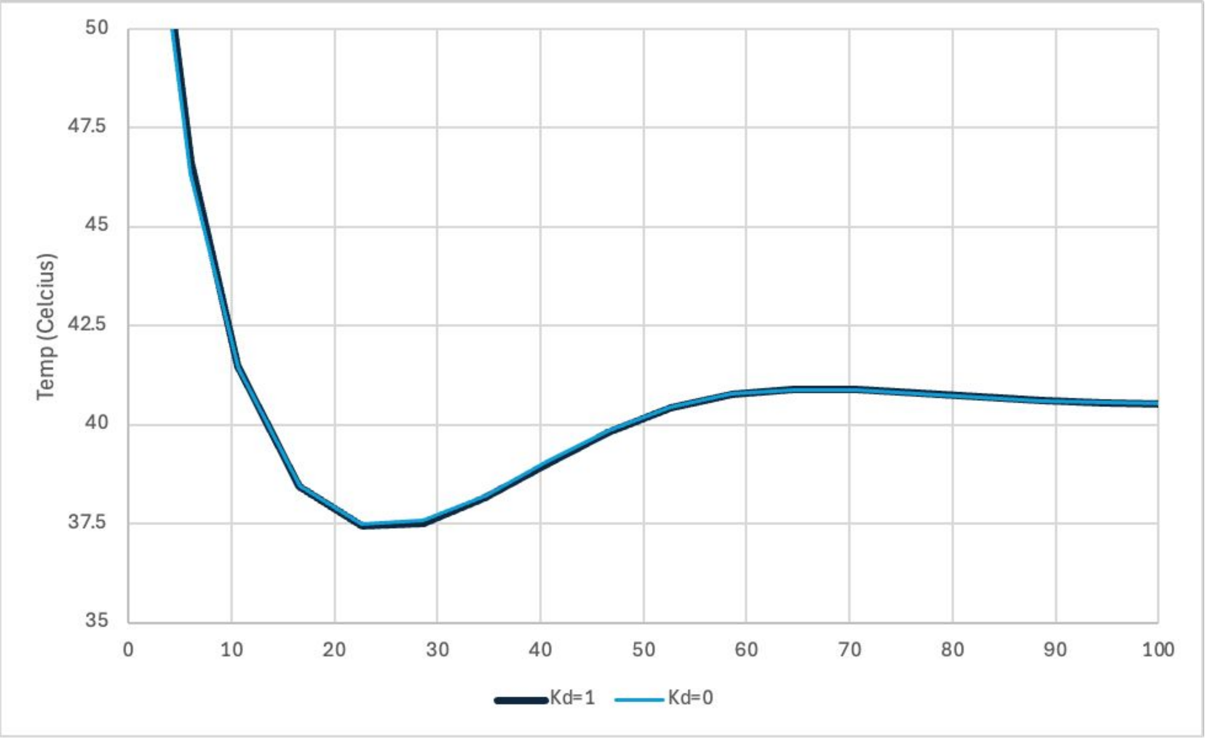
Optimal K_I : 1

The MATLAB simulation showed that smaller K_D values create a smoother approach to the setpoint.



Optimal K_D : 0

The MATLAB simulation showed that smaller K_D values create a smoother approach to the setpoint.



Optimal K_D : 0

Conclusion:

Automatic control can give superior control over the process when utilized properly. When poorly utilized, manual control can give better results.

Method	$T_{mix, theory}$ (avg)	$T_{mix, experimental}$ (avg)	% Difference (avg)
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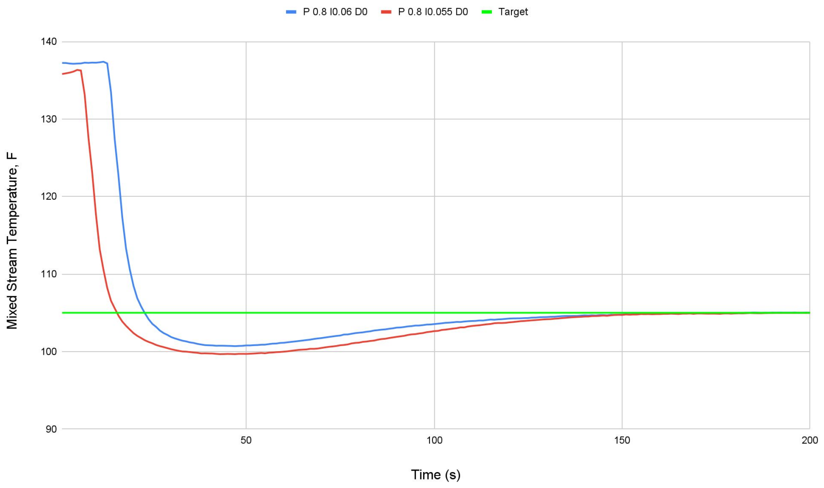

Questions?

References

[1] M. Cooper, P. Lim, K. Dickey, and K. Dickey, “Temperature Control Experiment” CHE 331 Course Handout, North Carolina State University, Raleigh, NC.

[2] O’BRIEN, TERRY. n.d. “Which Is Better: Automatic or Manual Internal Controls? | Schellman.” Schellman Compliance. https://www.schellman.com/blog/soc-examinations/automated-or-manual-soc-controls.

[3] “Feedback Control System - an Overview | ScienceDirect Topics.” n.d. Www.sciencedirect.com. https://www.sciencedirect.com/topics/engineering/feedback-control-system.